

Product Rule Notes

Howdy!

I will start with a review of things you have seen previously.

We know how to take derivatives of polynomial functions such as

$$y = x^4 + 3x^2 + 5x + 1$$

and maybe you have also seen

$$f(x) = e^x$$

before. But what happens if we combine them?

For instance, consider the function

$$f(x) = (x^2 + 67)e^x \tag{1}$$

or

$$y = (3x^2 + x + 1)(2x^3 + x) \tag{2}$$

which are made up of functions we know how to take derivatives of, but so far we don't know how to handle.

In both of these cases, we can apply something called the product rule. This applies any time we can write

$$f(x) = g(x) \cdot h(x).$$

Theorem.

$$f'(x) = g'(x)h(x) + g(x)h'(x).$$

In Example 1, we let

$$g(x) = x^2 + 67 \quad h(x) = e^x$$

so that

$$f(x) = g(x) \cdot h(x) = (x^2 + 67)e^x.$$

Then

$$g'(x) = 2x \quad h'(x) = e^x,$$

and together

$$f'(x) = g'(x)h(x) + g(x)h'(x) = 2xe^x + (x^2 + 67)e^x.$$

Next, what happens if we have three functions,

$$f(x) = a(x)b(x)c(x)?$$

Answer: It doesn't matter.